LC29H (AA, BA, CA, DA)&LC79H (AL)

OSNMA Application Note

GNSS Products

Version: 1.1

Date: 2026-09-23

Status: Released

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.
No. 8 Waipojing Road, Sijing Town, Songjiang District, Shanghai 201601, China
Tel: +86 21 5108 6236
Email: info@quectel.com

Or our local offices. For more information, please visit:
https://www.quectel.com/contact/.

For technical support, or to report documentation errors, please visit:
https://www.quectel.com/tech-support/.
Or email us at: support@quectel.com.

Legal Notices

We provide this document to support your product design. You are required to design your products based on the specifications and parameters set forth herein. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an “as available” basis. You acknowledge and agree that we may add to, amend, or restate this document at any time at our sole discretion without any prior notice to you, and such additions, amendments, or restatements shall be binding upon you.

Use and Disclosure Restrictions

License Agreements

The recipient of any hardware, software, materials, or documentation provided by us shall keep such content confidential, unless expressly authorized by us. The recipient shall not disclose, access, or use any part of the received content for any purpose other than the execution and implementation of the intended project.

Copyright

Our and third-party products hereunder may contain copyrighted materials, including but not limited to protected content, hardware, software, and documentation owned by us or applicable third parties. Unless prior written consent is obtained, you shall not access, use, or disclose any documents or information provided by us, nor shall you copy, reproduce, republish, display, translate, distribute, merge, modify, or create derivative works from any such copyrighted materials. We and the applicable third party retain exclusive rights to all copyrighted materials. No license to any patents, copyrights, trademarks, or service marks shall be granted or transferred. For the avoidance of doubt, no form of purchase shall be construed as granting any license beyond a normal, non-exclusive, royalty-free license to use the product. We reserve the right to pursue legal action against any violation of confidentiality obligations,

unauthorized use, or any other unlawful or malicious use of the aforementioned documents and information.

Trademarks

Unless otherwise expressly provided, nothing in this document shall be construed as conferring any rights to use any trademark, trade name, name, abbreviation, or counterfeit thereof owned by us or any third party in advertising, publicity, or any other contexts.

Third-Party Rights

You understand that this document may refer to hardware, software, and/or documentation owned by one or more third parties ("third-party materials"). Use of such third-party materials is subject to all applicable restrictions and obligations set forth herein.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing, course of performance, or usage of trade.

Privacy Policy

To enable product functionality, certain device data may be uploaded to our or third-party servers, including those operated by carriers, chipset suppliers, or servers designated by you. We strictly comply with applicable laws and regulations and will retain, use, disclose, or otherwise process relevant data solely for the purpose of enabling product functionality, or as permitted by applicable laws. Before interacting with any third party regarding data exchange, please be informed of and understand their privacy and data security policies.

Disclaimer

a) We shall not be liable for any damages resulting from failure to comply with applicable operational or design specifications.
b) We shall bear no liability for any inaccuracies or omissions in this document, nor for any damages arising from the use of the information contained herein.
c) While we have made every effort to ensure that the functions and features under development are free from errors, it is possible that they could contain errors, inaccuracies, and omissions. Unless otherwise provided by valid agreement, we make no warranties of any kind, either implied or express, and exclude all liability for any loss or damage suffered in connection with the use of features and functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage may have been foreseeable.
d) We are not responsible for the accessibility, safety, accuracy, availability, legality, or completeness of information, advertising, commercial offers, products, services, and materials on third-party websites and third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2026. All rights reserved.

About the Document

Document Information	
Title	**LC29H (AA, BA, CA, DA)&LC79H (AL) OSNMA Application Note**
Subtitle	GNSS Products
Document Type	Application Note
Document Status	Released

Revision History

Version	Date	Description
-	2022-04-15	Creation of the document
1.0	2025-10-22	First official release
1.1	2026-09-23	1. Added applicable modules LC29H (BA, CA, DA) and LC79H (AL). 2. Added **PQTMCFGNMAPAR2** and updated the example (*Chapter 2.2*).

Contents

About the Document **3**
Contents **4**
Table Index **5**
Figure Index **6**

1 Introduction **7**
1.1. OSNMA Overview 7
1.2. OSNMA Data 7
1.2.1. Tag 7
1.2.2. TESLA Chain Key 7
1.2.3. TESLA Root Key 8
1.2.4. Public Key 8
1.2.5. Merkle Tree Root 8
1.3. OSNMA Processing Logic 9
1.4. OSNMA Applications 10

2 OSNMA Operation Guide **13**
2.1. Obtaining Public Key and Merkle Tree Root 13
2.2. Module Configuration 15

3 FAQs **18**

4 Appendix References **19**

Table Index

Table 1: Related Documents 19
Table 2: Terms and Abbreviations 19

Figure Index

Figure 1: OSNMA Processing Logic 10
Figure 2: Vehicle Navigation 11
Figure 3: Maritime Transport 11
Figure 4: Drone Safe Navigation 12
Figure 5: Geological Survey 12
Figure 6: Log in or Register an Account on GSC Website 13
Figure 7: OSNMA_MERKLETREE and OSNMA_PUBLICKEY Options 14
Figure 8: Retrieve Merkle Tree Root Information 14
Figure 9: Retrieve Public Key Information 15

1 Introduction

This document describes the OSNMA feature of Quectel LC29H (AA, BA, CA, DA) and LC79H (AL) GNSS modules, including OSNMA data, processing logic, application scenarios, operation guide, and FAQs.

1.1. OSNMA Overview

The OSNMA (Open Service Navigation Message Authentication) feature of the module is a technology used to enhance the security and reliability of GNSS receiver. The OSNMA is based on the navigation message authentication architecture. Galileo satellite issues MAC tags and digitally signed TESLA root key. After verifying the digital signature and tags, the module can identify tampered and forged navigation messages, enhancing the anti-spoofing capability of the system. OSNMA is a free, global Galileo navigation message authentication service, with authentication data transmitted via E1-B I/NAV navigation messages.

1.2. OSNMA Data

1.2.1. Tag

- Purpose:
  Tag is broadcast with navigation messages by the Galileo satellites. The module uses the navigation data and the TESLA chain key to calculate a tag, and then compares this tag with the broadcast tag. If two tags match, it indicates the navigation message is intact during transmission and comes from the genuine Galileo satellites. The module will use this navigation message for positioning and timing.

- Source:
  Galileo navigation message.

1.2.2. TESLA Chain Key

- Purpose:
  Galileo satellites broadcast the next transmitted TESLA chain keys. If the root key of the whole TESLA key chain has been verified, the module can reconstruct and verify the preceding TESLA

chain keys from the received next transmitted TESLA chain keys via the one-way hash function, subsequently verifying the authentication tag.

- Source:
  Galileo navigation message.

1.2.3. TESLA Root Key

- Purpose:
  TESLA root key verifies the authenticity and integrity of the TESLA chain key. The module cannot directly trust the broadcast TESLA root key since attackers can also forge a root key. To solve this issue, Galileo system uses the digital signature technology. After receiving the TESLA root key, the module verifies the digital signature using the public key obtained from the European GNSS Service Centre (GSC) or broadcast by Galileo satellites.

- Source:
  Galileo navigation message.

1.2.4. Public Key

- Purpose:
  Public key verifies the digital signature of the TESLA root key, thereby verifying that the TESLA root key data obtained by the user is authentic and has not been tampered.

- Source:
  GSC or broadcast in the Galileo navigation message. The public key can be obtained from the Galileo navigation message at specific times (GST 00:00, 06:00, 12:00, and 18:00).

1.2.5. Merkle Tree Root

- Purpose:
  The Merkle tree root verifies the public key issued by the satellite navigation message and confirms that the public key retrieved from the SIS is real and has not been tampered.

- Source:
  GSC.

1.3. OSNMA Processing Logic

The OSNMA authentication process is a cryptographic authentication mechanism based on the TESLA protocol. See core process as follows:

1. Signal generation and broadcasting (space side)
   For each specified time interval *i*, the Galileo satellites will:
   - Generate a navigation message (NAV message) containing navigation data.
   - Calculate an authentication tag for the navigation message using the unpublished TESLA chain key (*Ki*) corresponding to the current time interval and a cryptographic hash function (such as SHA-256).
   - Broadcast a data set: Navigation message, the delayed-disclosed TESLA chain key (*Kj, j* < *i*) used to verify the previously generated authentication tags, the authentication tag and the TESLA root key.

2. Signal reception and preliminary verification (module side)
   After receiving the signal, the module that supports OSNMA will:
   - Parse the signal to obtain the navigation message, the published TESLA chain key, the authentication tag, and the TESLA root key.
   - Verify the TESLA root key: The module cannot directly trust the broadcast TESLA root key since attackers can also forge a root key. To solve this issue, Galileo system uses the digital signature technology. After receiving the TESLA root key, the module verifies it via its digital signature using a public key obtained from the GSC or broadcast by Galileo satellites.
   - Verify the TESLA chain key: After the root key verification is completed, the module proceeds to verify the chain key. Using the TESLA root key verified with the public key as the trust anchor, the module recursively derives the preceding chain keys from the next transmitted TESLA chain keys broadcast by the satellite, using the SHA-256 one-way hash function. If the result by multiple hash operations matches the trusted root key, the received chain key is verified as valid, thereby establishing trust in the TESLA key chain.

3. Reconstruction and authentication (module side)
   After verifying the chain key, for the final authentication the module will:
   - Verify the validity of the authentication key: Verify the validity of the key chain via the verified chain key *Kj* and one-way hash function.
   - Compute the tag locally: Using the verified TESLA chain key *Kj* and the locally cached navigation message from the previous time slot as inputs, the module locally regenerates the corresponding authentication tag with the same HMAC algorithm used by the satellite.
   - Comparison: Finally, the receiver compares the locally computed tag bit by bit with the tag received and cached from the previous time slot. If the tags match, the navigation data from the previous time slot is authenticated.

4. Authentication result
   - Match: The cached navigation message from the previous time slot passes the verification,

indicating that the navigation data was transmitted completely and without tampering and originated from a legitimate Galileo satellite. The module can use the navigation information for positioning and timing.

- Mismatch: There are two possible causes. First, the navigation message may have been corrupted during transmission or the received data may be incomplete. Second, the data may have been tampered with or the receiver may have received spoofed satellite signals. The module discards the navigation data and, if necessary, outputs an anomaly alert.

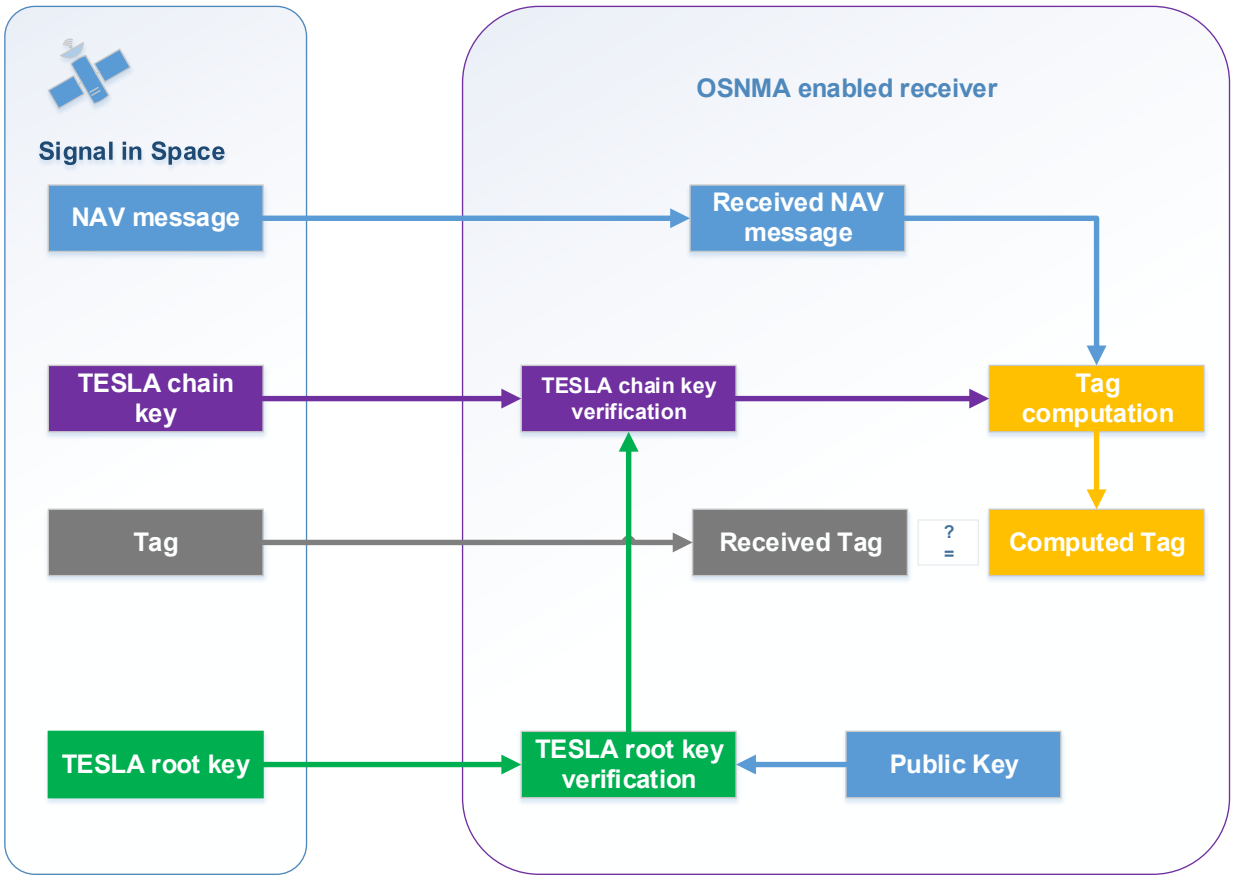


Figure 1: OSNMA Processing Logic

1.4. OSNMA Applications

As a GNSS anti-spoofing technology, OSNMA has broad applications in various vertical industries that use GNSS signals for positioning, navigation or timing. By providing authenticated navigation messages, OSNMA feature ensures accurate positioning, navigation or timing for various GNSS module application scenarios.



Figure 2: Vehicle Navigation



Figure 3: Maritime Transport



Figure 4: Drone Safe Navigation



Figure 5: Geological Survey

2 OSNMA Operation Guide

2.1. Obtaining Public Key and Merkle Tree Root

Users can log in to the GSC to obtain valid public key and Merkle tree root information.

Step 1: Log in or register your account on the GSC website: https://www.gsc-europa.eu/gsc-products/pki.

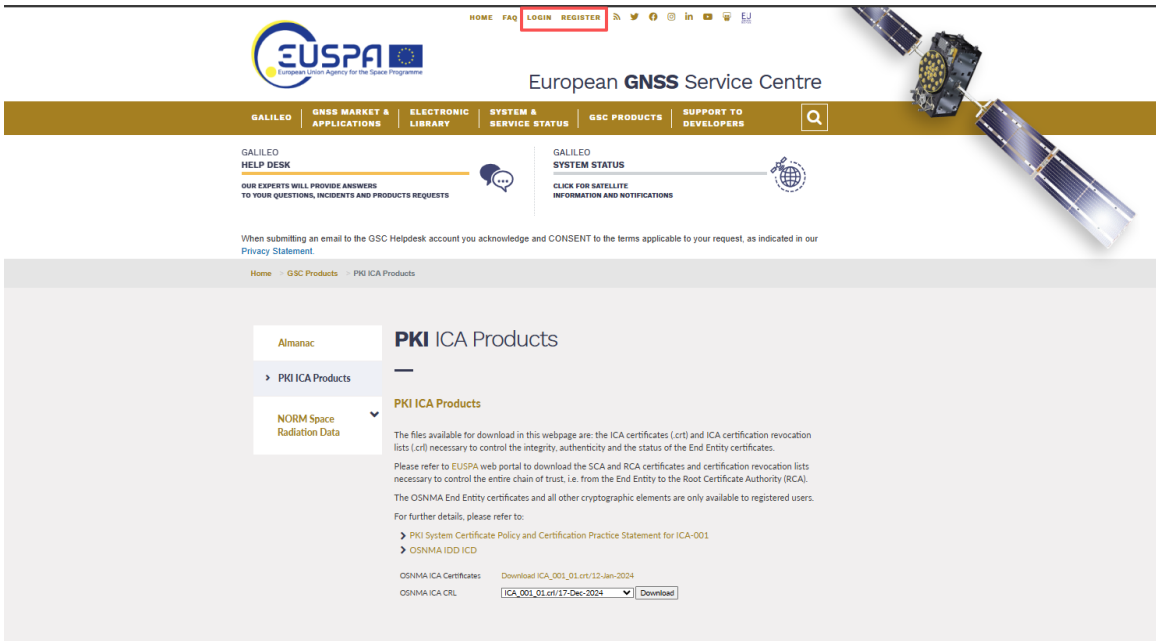


Figure 6: Log in or Register an Account on GSC Website

Step 2: Retrieve the public key and Merkle tree root information. After a successful registration, log in to your account and click **"GSC PRODUCTS"** to view the available options, namely OSNMA_MERKLETREE and OSNMA_PUBLICKEY.

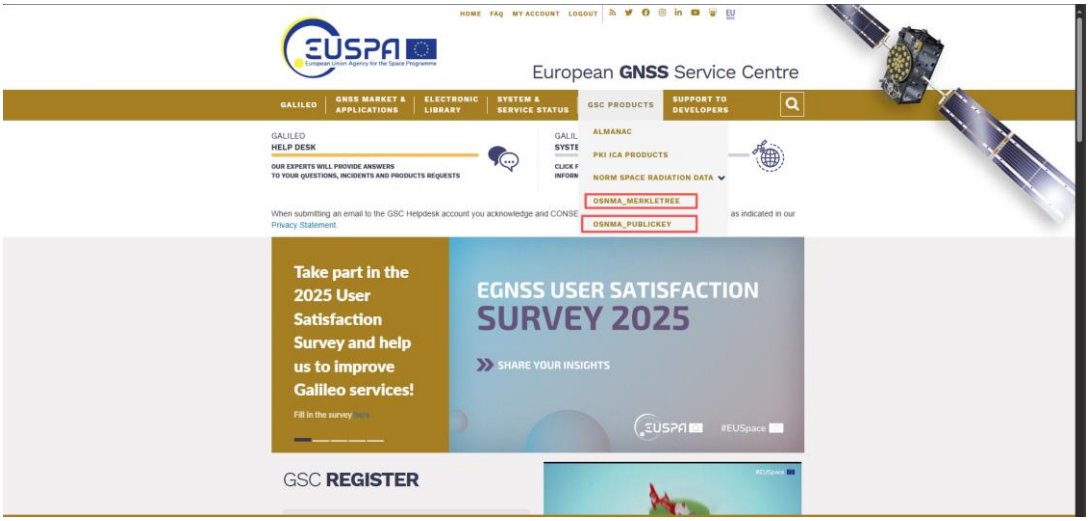


Figure 7: OSNMA_MERKLETREE and OSNMA_PUBLICKEY Options

Step 3: Click "**OSNMA_MERKLETREE**" to retrieve the Merkle tree root information.



Figure 8: Retrieve Merkle Tree Root Information

Step 4: Click "**OSNMA_PUBLICKEY**" to retrieve the public key information.

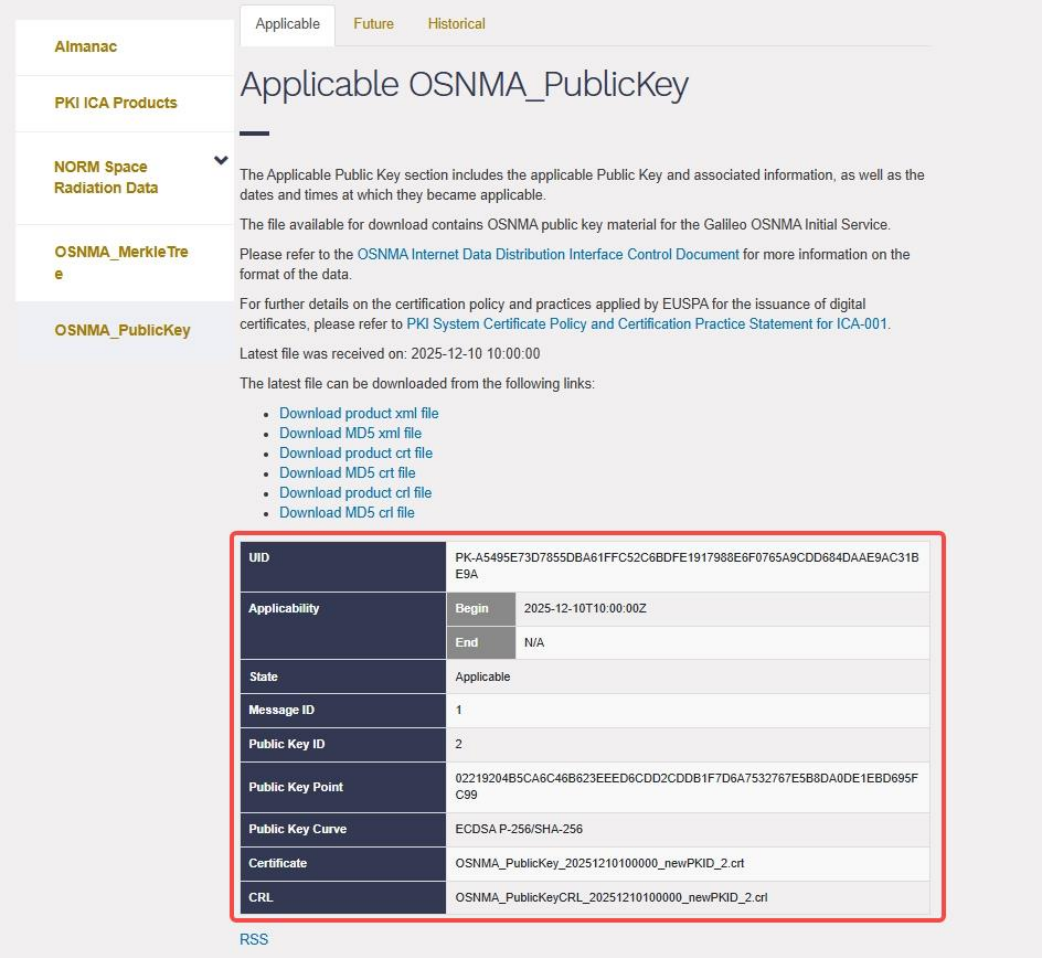


Figure 9: Retrieve Public Key Information

NOTE

The module firmware contains the latest default public key and Merkle tree root released by the GSC. For the firmware version supporting OSNMA feature, please contact Quectel Technical Support.

2.2. Module Configuration

For details of the PQTM commands related to OSNMA in the following steps, please refer to *document [1] protocol specification*.

Step 1: Power on the module, open the QGNSS tool, and select the module and its corresponding COM port. For specific usage on QGNSS tool, see *document [2] QGNSS user guide*.

Step 2: Check whether the module output and positioning status are normal, and whether it can track the Galileo satellite normally.

Step 3: The OSNMA feature is disabled by default. You must execute **PQTMCFGNMA** to enable the OSNMA feature.

Step 4: Set the Merkle tree root for the GNSS module via **PQTMCFGNMAPAR** or **PQTMCFGNMAPAR2**. A Merkle tree root is preloaded to the module by default, so you can omit this step.

Step 5: Set the public key for the GNSS module via **PQTMCFGNMAPAR** or **PQTMCFGNMAPAR2**. The public key is preloaded to the module by default, so you can omit this step.

Step 6: Enable the output of **PQTMNMASTATUS**, **AMC**, and **ASA** messages via **PQTMCFGMSGRATE**; then execute **$PQTMSAVEPAR*5A** to save the configuration, and restart the module.

Step 7: Input the reference UTC time for OSNMA time synchronization via **PQTMNMATIMESYNC**. The reference time must be input after the module restarts. The accuracy of the input reference time determines whether the module enters OSNMA Fast MAC mode or Slow MAC mode.

Step 8: After a successful authentication, the **AMC** outputs the authenticated position, time, date and speed data, the **ASA** displays the authenticated satellite ID in use and the value of **<OSNMA_Status>** in **PQTMNMASTATUS** switches to 1.

Example:

```
//Enable OSNMA feature.
$PQTMCFGNMA,W,1,0,0,0,0,0*62
$PQTMCFGNMA,OK*30
//Set the Merkle tree root using PQTMCFGNMAPAR.
$PQTMCFGNMAPAR,W,0,2,0,4,0,256,832E15EDE55655EAC6E399A539477B7C034CCE24C3C93FFC904ACD9BF842F04E*0B
$PQTMCFGNMAPAR,OK,0,2*71
//Or set the Merkle tree root using PQTMCFGNMAPAR2.
$PQTMCFGNMAPAR2,W,0,2,0,4,0,256,832E15EDE55655EAC6E399A539477B7C,034CCE24C3C93FFC904ACD9BF842F04E*15
$PQTMCFGNMAPAR2,OK,0,2*43
//Set the public key using PQTMCFGNMAPAR.
$PQTMCFGNMAPAR,W,0,1,2,1,33,02219204B5CA6C46B623EEED6CDD2CDDB1F7D6A7532767E5B8DA0DE1EBD695FC99*29
$PQTMCFGNMAPAR,OK,0,1*72
//Or set the public key using PQTMCFGNMAPAR2.
$PQTMCFGNMAPAR2,W,0,1,2,1,33,02219204B5CA6C46B623EEED6CDD2CDD,B1F7D6A7532767E5B8DA0DE1EBD695FC,99*1B
```

$PQTMCFGNMAPAR2,OK,0,1*40
//Enable the output of **PQTMNMASTATUS**.
$PQTMCFGMSGRATE,W,PQTMNMASTATUS,1,1*18
$PQTMCFGMSGRATE,OK*29
//Enable the output of **AMC**.
$PQTMCFGMSGRATE,W,AMC,1*04
$PQTMCFGMSGRATE,OK*29
//Enable the output of **ASA**.
$PQTMCFGMSGRATE,W,ASA,1*18
$PQTMCFGMSGRATE,OK*29
//Save the configuration.
$PQTMSAVEPAR*5A
//After the module restarts, input the reference UTC time for OSNMA time synchronization.
$PQTMNMATIMESYNC,W,2026,04,23,12,37,10,18,3*0C
$PQTMNMATIMESYNC,OK*60
//**AMC** outputs the authenticated position, time, date and speed data.
$GNAMC,124344.000,A,2516.137791,N,11020.035637,E,0.02,,230426,,,A,V*0C
//**ASA** displays the authenticated satellite ID in use.
$GNASA,A,3,11,10,04,09,12,36,23,31,,,,,1.85,1.31,1.30,3*01
//After a successful authentication, the value of **<OSNMA_Status>** in **PQTMNMASTATUS** switches to 1.
$PQTMNMASTATUS,1,20260423,124344.000,25.26896320,110.33392730,188.815,4,1,,,,,*55

NOTE

1. The **PQTMCFGNMAPAR** command is supported only on LC29H (AA) and LC79H (AL).
2. The **PQTMCFGNMAPAR2** command is supported only on LC29H (BA, CA, DA). The LC29H (CA) module with firmware versions dedicated to two-wheeled vehicles currently does not support the **PQTMCFGNMAPAR2** command.

3 FAQs

1. **Does each GNSS receiver have a specific public key?**

   During the validity period of the public key, all receivers use the same public key for authentication. This public key is published by the GSC or broadcast in the Galileo navigation message.

2. **Do public key and Merkle tree root have to be set up manually?**

   The public key and Merkle tree root in the module firmware have default values. The default values are the latest public key and Merkle tree root at the time of release. Generally, there is no need to manually configure the public key and Merkle tree root. The public key can be retrieved from the Galileo navigation message at a specific time (GST 00:00, 06:00, 12:00, 18:00). Merkle tree root cannot be retrieved from satellite navigation messages. It can only be manually configured in the module.

3. **What are the similarities and differences between the fast MAC and slow MAC authentication mechanisms in the OSNMA feature?**
   1) Same data type. Both Fast MAC and Slow MAC are used to authenticate ephemeris data.
   2) Different time accuracy requirements. To ensure verification reliability, Fast MAC requires the time difference between the input time and the received Galileo broadcast time to be within 15 seconds, whereas Slow MAC requires the time difference to be within 165 seconds.
   3) Different TESLA chain keys. Fast MAC uses the TESLA chain key published in the first subframe immediately following the current subframe to verify the data in the current subframe. Therefore, Fast MAC can verify data within 30 seconds. Slow MAC uses the TESLA chain key published in the 10th subframe after the current subframe to verify the data of the current subframe. This results in a delay of about 5 minutes (10 subframes × 30 seconds/subframe = 300 seconds). Therefore, Slow MAC requires at least 5 minutes for successful verification.

4 Appendix References

Table 1: Related Documents

Document Name
[1] Quectel_LC29H&LC79H_Series_GNSS_Protocol_Specification
[2] Quectel_QGNSS_User_Guide

Table 2: Terms and Abbreviations

Abbreviation	Description
Galileo	Galileo Satellite Navigation System (EU)
GNSS	Global Navigation Satellite System
GSC	European GNSS Service Centre
GST	Galileo System Time
HMAC	Hash Message Authentication Code
MAC	Message Authentication Code
OSNMA	Open Service Navigation Message Authentication
PQTM	Quectel Proprietary Message
SHA	Secure Hash Algorithm
SIS	Signal in Space
TESLA	Timed Efficient Stream Loss-tolerant Authentication
UTC	Coordinated Universal Time